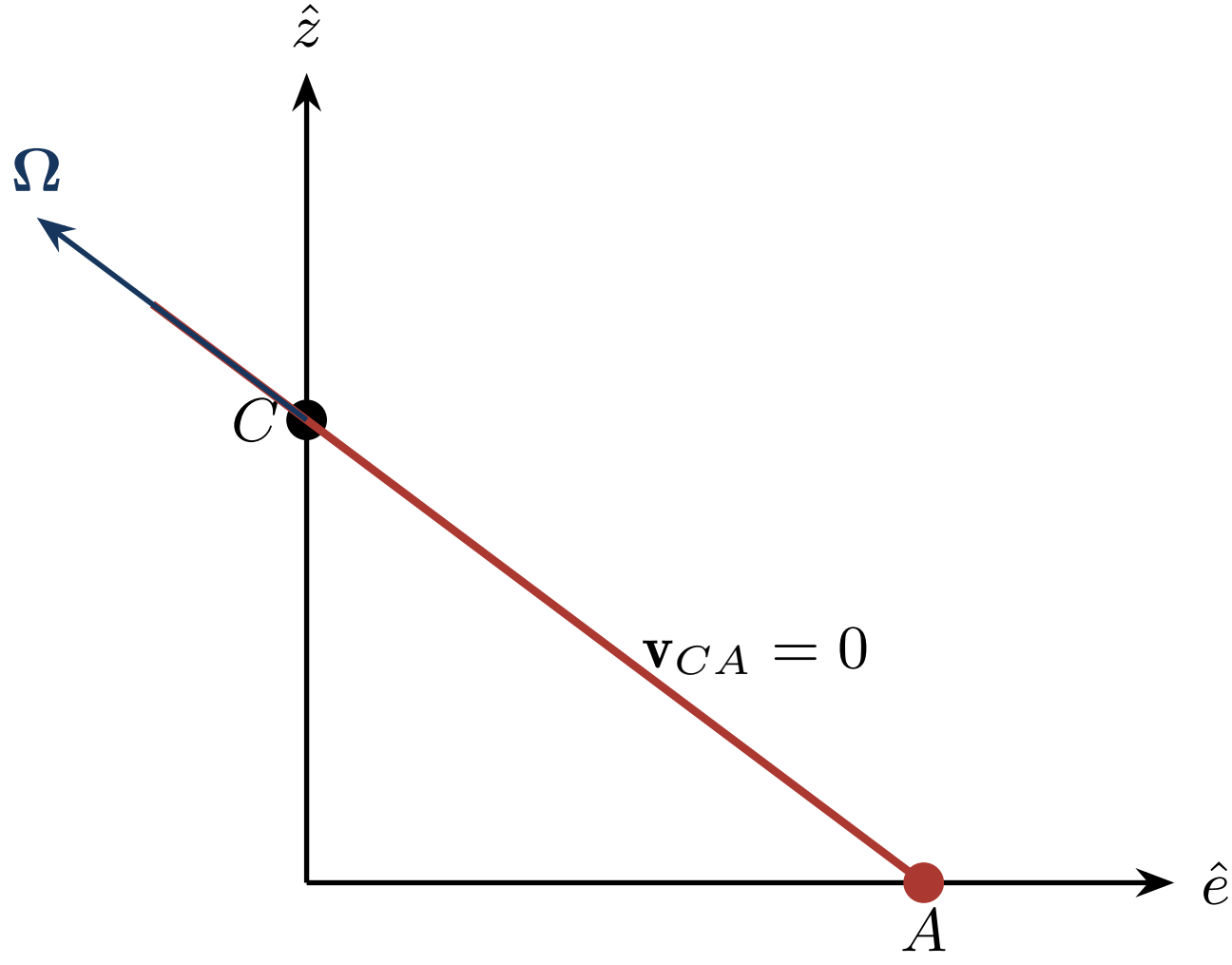

$$\boldsymbol{\Omega} = -\omega \sin\theta \, \hat{u}$$

$$|\boldsymbol{\Omega}| = |\omega| \sin\theta$$